

FULL-STACK DEVELOPMENT · MULTI-LEVEL

Full-Stack Developer Path

Build the entire application, not just one layer.



Frontend foundations to MERN, MEAN, .NET or Java backend specialization.

🕒 16-24 weeks

📁 4 modules

📖 16 lessons

[Explore the course →](#)

Concepts, worked examples, assignments and a coherent course project.

WHO IT IS FOR · WHAT YOU WILL LEARN

Your next capability.

Build accessible web interfaces, secure APIs, data-backed applications, automated tests and a production-ready capstone.



Build responsive web applications



Develop secure production APIs



Work with SQL and NoSQL data



Ship a portfolio-grade capstone



Audience

- Students
- Developers switching stacks
- Career returners



Prerequisites

- No framework experience required

FROM CONCEPT TO EVIDENCE



One curriculum, three connected views

This guide summarizes the same modules and lessons shown on the course overview and in the learning workspace. The next pages explain the scope and practical evidence for every module.

DETAILED MODULE CURRICULUM · 1-2

Build the foundations.

The module names, lesson sequence and evidence below match the course learning workspace.

1 Web Foundations

Build a semantic, responsive profile page before adding a framework.

1.1 Semantic HTML

HTML describes document meaning.

1.2 Modern CSS

CSS separates presentation from structure.

1.3 JavaScript and TypeScript

JavaScript provides runtime behavior; TypeScript adds static checks during development.

1.4 Accessibility and responsive design

Responsive design adapts content and controls, not just font size.

You will produce: Responsive page and accessibility checklist.

2 Frontend Engineering

Organize a frontend around components, explicit state and testable behavior.

2.1 React or Angular foundations

A component has a focused responsibility and a clear input contract.

2.2 State and forms

State should represent the minimum information needed to render the interface.

2.3 Data fetching

Data fetching introduces loading, failure, empty and stale states.

2.4 Frontend testing

Test components through observable behavior.

You will produce: Component interface, state diagram and UI tests.

Practice is part of every module

Work through the example, record your reasoning, complete the module assignment and compare your work with the review criteria.

DETAILED MODULE CURRICULUM · 3-4

Apply, integrate and demonstrate.

The module names, lesson sequence and evidence below match the course learning workspace.

3 Backend Specialization

Create an API that enforces rules independently of its user interface.

3.1 API design

An API contract defines resource identity, validation, response shape and errors.

3.2 Authentication and authorization

Authentication establishes identity; authorization evaluates whether that identity may perform an operation on a resource.

3.3 Validation and errors

Validate types, ranges and required relationships at the boundary.

3.4 Backend testing

Backend tests should exercise contracts, permissions and state transitions.

You will produce: API specification and permission regression tests.

4 Data and Production

Connect data modeling, release practice and the final application.

4.1 SQL modeling

SQL modeling defines keys, relationships and constraints that protect data integrity.

4.2 NoSQL patterns

Document-oriented data models trade joins for embedded or referenced structures.

4.3 Docker and CI/CD

A container packages an application environment; CI verifies and builds changes.

4.4 Full-stack capstone

The capstone integrates a responsive frontend, validated API, data model and tests.

You will produce: Full-stack capstone and deployment runbook.

Practice is part of every module

Work through the example, record your reasoning, complete the module assignment and compare your work with the review criteria.

PROJECTS AND WORKING METHODS

A complete web application

Build a responsive interface, connect an API and database, implement access rules and prepare a tested deployment.

**Responsive user interface****API and data model****Tested release and project
README****Tools and methods**

HTML, CSS and TypeScript

React or Angular

Node.js, .NET or Java

SQL and MongoDB

Git, testing and cloud

Tools are learning choices, not partner endorsements.

Inside module 1: a worked example

```
<main>
  <article aria-labelledby="profile-title">
    <h1 id="profile-title">Learner profile</h1>
    <p>My current learning project.</p>
    <a href="./projects.html">View projects</a>
  </article>
</main>
```

The link navigates to another resource; it should not be replaced with a button merely to achieve a visual style.

How your work is reviewed

Show the artifact, explain your decisions, test the stated assumptions and identify limitations. Module assignments build toward the course project rather than replacing it with a single example.

DELIVERY, COMPLETION AND NEXT STEPS

Know what you are signing up for.

**Delivery**

Hybrid learning · 16–24 weeks

Confirm current schedule, cohort arrangements and enrollment terms with the academy.

**Completion requirements**

Complete the required coursework and module assignments, and pass the course assessment.

A course completion certificate is available after those requirements are met.

What the learning workspace includes**Follow the modules**

Expand the curriculum and open the exact lesson you need.

**Apply the concepts**

Read explanations, inspect examples and complete module practice.

**Keep your evidence**

Use resources, notes, discussion and assignments to support your learning.

Learning commitments, not outcome guarantees

Completion is not a placement, employment or performance guarantee. The course project is educational evidence; any production use requires appropriate review and validation.

Start with the curriculum. Continue with the first lesson.

Review your starting point, download this guide and explore the connected course experience.

[Go to the first lesson](#) →

Client-review edition. The static learning workspace saves demonstration progress locally; production enrollment, assessment, communications and certificate issuance are not connected.